Serial Decode — user manual

Ian Ameline · version 1.42 · MIT licence

This app turns a Keithley bench instrument into a serial decoder. Clip onto a UART line, press **Capture**, and read the bytes on the front panel. You do not have to tell it the baud rate, the frame format or which way up the signal is — it works all of that out from the signal itself.

It was written on a **DMM6500**, and that is **the only instrument it has been tested on**. It should run as it is on the rest of the Keithley TSP range that has a digitizer and the touchscreen app API — the DAQ6510 and DMM7510 in particular — and it will install on an SMU2461 and try. **Which instruments** in the README says what is measured and what is inference.

Three things make it worth reaching for over a scope's decode: the **window is large** — a screenful is ~240 bytes and a recording holds 32 768 — every capture is **appended to the USB key automatically** while a key is in the slot, and the **serial parameters are detected rather than configured**. Connect, press, read.

> **Every number in this manual was measured on a real DMM6500** against a real signal generator unless the text says otherwise; anything calculated or untested says so where it appears. What that cannot cover is the variety of real devices, so if the app tells you something you can prove is wrong, that is the bug worth reporting. It is built so that **refusing with a reason is acceptable and confident nonsense is not**, and anything it is unsure of it says on the note row at the bottom of the screen.

Hooking it up — read this part

The line must stay between −10 V and +10 V. The app fixes the meter on its 10 V range. 3.3 V CMOS, 5 V TTL and a 6 V LIN line tapped to ground are all fine as they are. For a 12 V line, put a 2:1 resistor divider in front of it — and if it is an open-drain bus, pick resistors big enough not to load it down.

Connect INPUT LO to the ground of the circuit you are probing. Everything is measured at INPUT HI relative to INPUT LO, so without a shared ground the readings mean nothing.

Do not put a differential bus straight across HI and LO. RS-485, CAN and LIN need to go through a transceiver first, or you need to tap one side of the pair against ground.

A DC offset does not matter. The app measures the two voltage levels on your line and puts its decision threshold between them, so it does not care where they sit — only that both fit inside ±10 V. A 1.6 V swing sitting up at 6 V decodes fine. **The tested range is a 0.33 V swing to an 8 V swing**, and signals spanning 9.3 V peak-to-peak decode when the extra span is spikes on a smaller logic swing. A 60 mV swing is refused outright.

A small swing needs a quiet line, not a special setting. The noise the app tolerates is a fraction of your swing, around 40 %, so 50 mV of noise is nothing on a 3.3 V line and a tenth of the budget on a 0.5 V one.

The app takes over the measurement function, and gives it back when you end it. Capturing means digitizing, so every press of Capture selects **Digitize Voltage** on the 10 V range — whatever the meter was measuring before. The mode you were in is read once when the app starts and put back when you press **End App**, so ending the app leaves the meter measuring what it was measuring when you launched it, on the range it was on.

Two things follow from that, both deliberate:

- **Changing the function from the front panel while the app is loaded does not stick.** The next Capture selects Digitize Voltage again — so the app keeps working, which is the point — and End App restores the mode from when you *launched*, not the one you switched to.
- **If you were already in Digitize Voltage when you launched**, the app puts your range, sample rate and aperture back too, since those are the settings it changes.

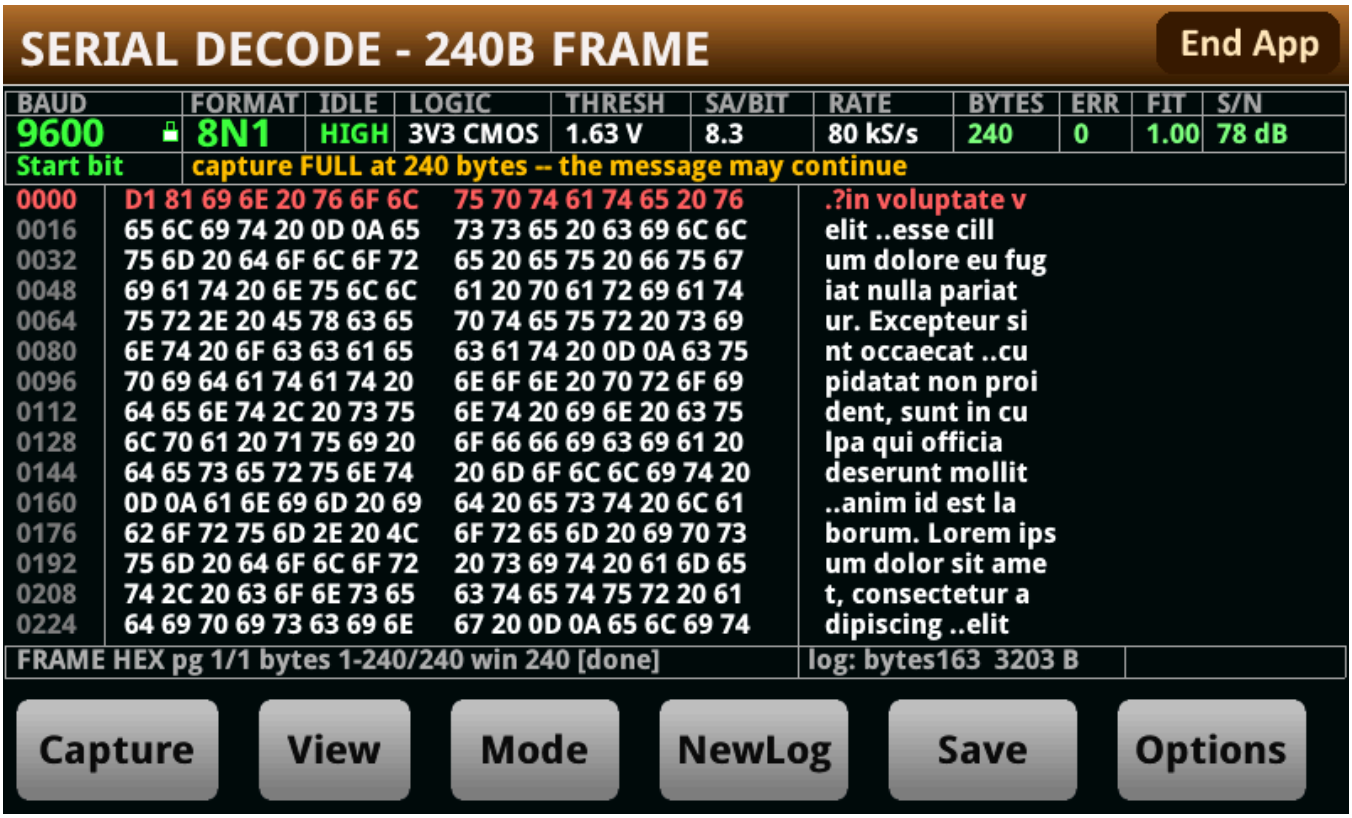
Quick start

1. Connect the line to **INPUT HI** and its ground to **INPUT LO**.
2. Press **Capture** while the device is talking.
3. Read the bytes. Press **View** to switch between hex and plain text.

That is the whole normal workflow. The first capture also locks the baud rate when it is confident, so later captures are faster.

Two more things worth knowing before you need them: **Mode** turns Capture into a recording of up to 32 768 bytes straight to the USB key, still one press — and once a long job starts, **nothing stops it early**, so decide the size before you press — see **The TRIGGER key** below.

The screen



Along the top are the things the app worked out about your line:

Field	Meaning
BAUD	The rate it found. A padlock means it is locked and will not be re-detected.

Field	Meaning
FORMAT	Data bits, parity, stop bits — `8N1`, `7E1` and so on.
IDLE	What the line rests at: `HIGH` for CMOS/TTL, `LOW` for an RS-232-style line.
LOGIC	The family it looks like, e.g. `3V3 CMOS`, `5V TTL`.
THRESH	The voltage it is using to tell 0 from 1.
SA/BIT	Samples per bit. More is better; 8 is comfortable, 4 still works.
RATE	How fast the meter sampled.
BYTES / ERR	How many bytes came out, and how many of them you cannot trust.
FIT	How well one bit-length explains what it measured, from 0 to 1. Below about 0.9 means the timing was messy.
S/N	How far your two logic levels stand above the noise sitting on them, in dB. Bigger is quieter.

A dash means "not measured yet".

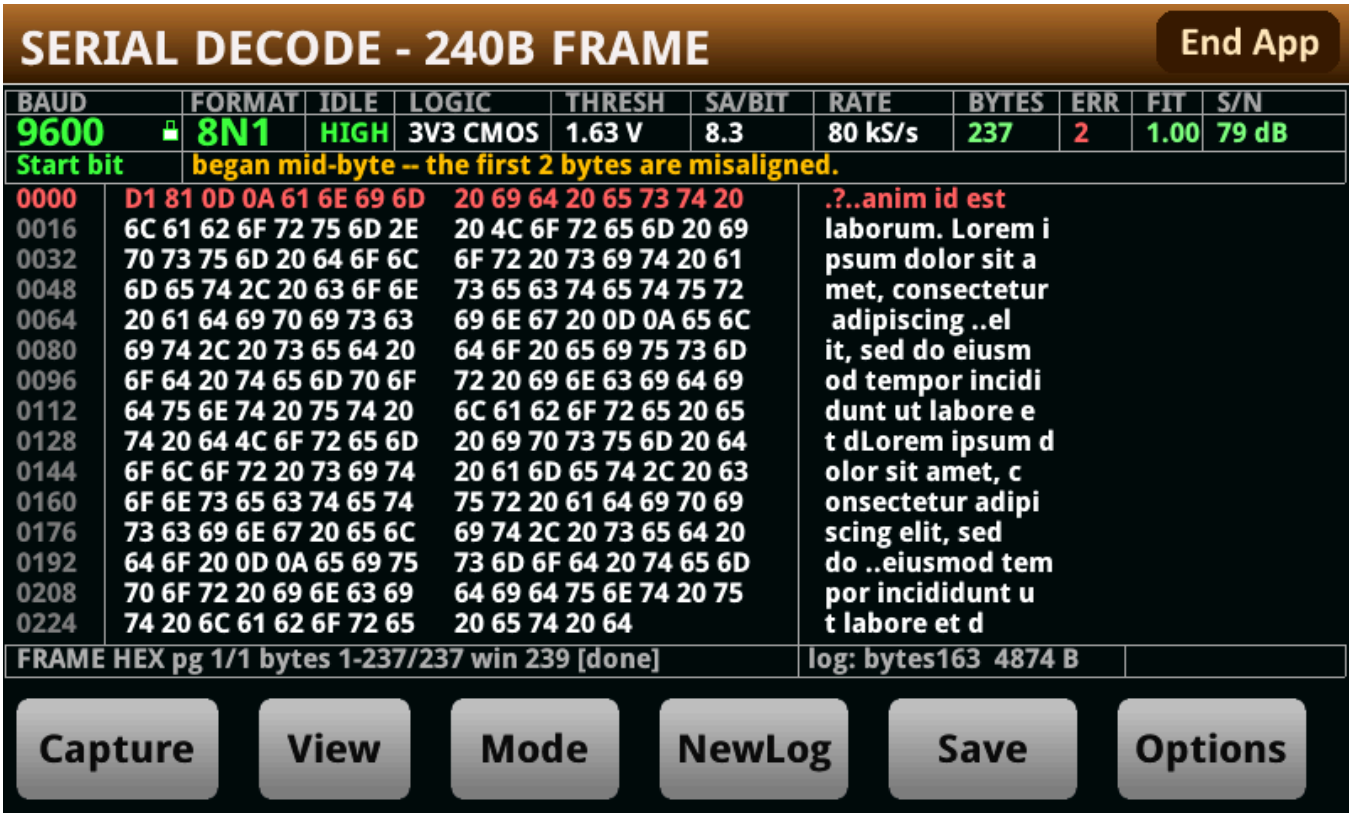
`FIT` and `S/N` are coloured, and the colour is the quick read. Green means there is margin, amber means there is some but not much, red means you are close to the edge — so the top line can be checked at a glance instead of remembering two sets of numbers:

	green	amber	red
FIT	0.90 and up	0.72 to 0.89	below 0.72
S/N	30 dB and up	15 to 29 dB	below 15 dB

`FIT` 0.90 is the same figure auto-lock requires, so a green `FIT` is also "this capture was good enough to lock". **Red means close to failing, not failing.** A red `S/N` still decodes: the point where captures stop working is around 12 dB, and past it the app says `no clear logic levels` rather than handing you wrong bytes. Green `S/N` is noise under about 5 % of your swing, amber down to about a third of it.

`S/N` describes your wiring, not the app. It is measured on the flat parts of the signal, away from the edges, so it reports what is riding on your levels — mains hum picked up by a long unshielded lead, a switching supply, a noisy ground. If it reads low, the wiring is worth fixing before anything else: every other tolerance in this manual is a fraction of the swing, so noise spends the same budget that jitter and slow edges do. The cell stops at `>99 dB` rather than counting higher: above that it would be reporting the meter's own digitizer resolution rather than anything about your line, so what it shows there is a limit and not a measurement.

Broken bytes are shown, not hidden. `ERR` turns red when there are any, and every row containing one is drawn red across its whole width — offset, hex and ASCII together — so damage is found by looking rather than by counting. Flagged bytes read as `?` in the ASCII gutter:



That is a real 240B capture that began part way through a byte. **The note row names the cause** — `began mid-byte -- the first N bytes are misaligned.` — which is the difference between a puzzle and a known condition: the app arms on a busy line, so it lands wherever the traffic happens to be, and the bytes before the first gap between messages cannot be framed. On a busy line roughly one capture in eight starts this way.

The note and `ERR` agree, and that is deliberate. The note's number is how far the damage reaches — the last byte in the opening region the decoder could not frame — so "the first N" is literally true, nothing past N is being accused, and `ERR` is never less than N. Above, both say 1. `ERR` counts the whole region rather than only the frames that failed a check: inside a misaligned head the bit boundaries are in the wrong places, so a frame there can satisfy its parity and its stop bit by luck and still hand back an ordinary character that is the wrong byte. **`ERR` counts bytes you cannot trust, not checks that failed.**

Two exclusions keep `ERR` at 0 on good data. Where no misaligned head is identified, the first three frames are ignored: a gapless line has no idle for the framer to anchor on, so it routinely resynchronises a frame or two in, and counting those would read `ERR 1` or `ERR 2` on every healthy capture. The final frame is ignored always, because the capture boundary cuts it in half. Neither exclusion applies across a head the app has evidence for.

Everything from the first gap onward decoded cleanly, and the log on the USB key holds every byte: one that failed its parity or stop bit is written and marked, not dropped.

Underneath: the **trigger cell**, the bytes, then two rows of text. The **status row** tells you where you are (`FRAME HEX pg 1/1 bytes 1–239/239 win 240 [done]`) and names the log file.

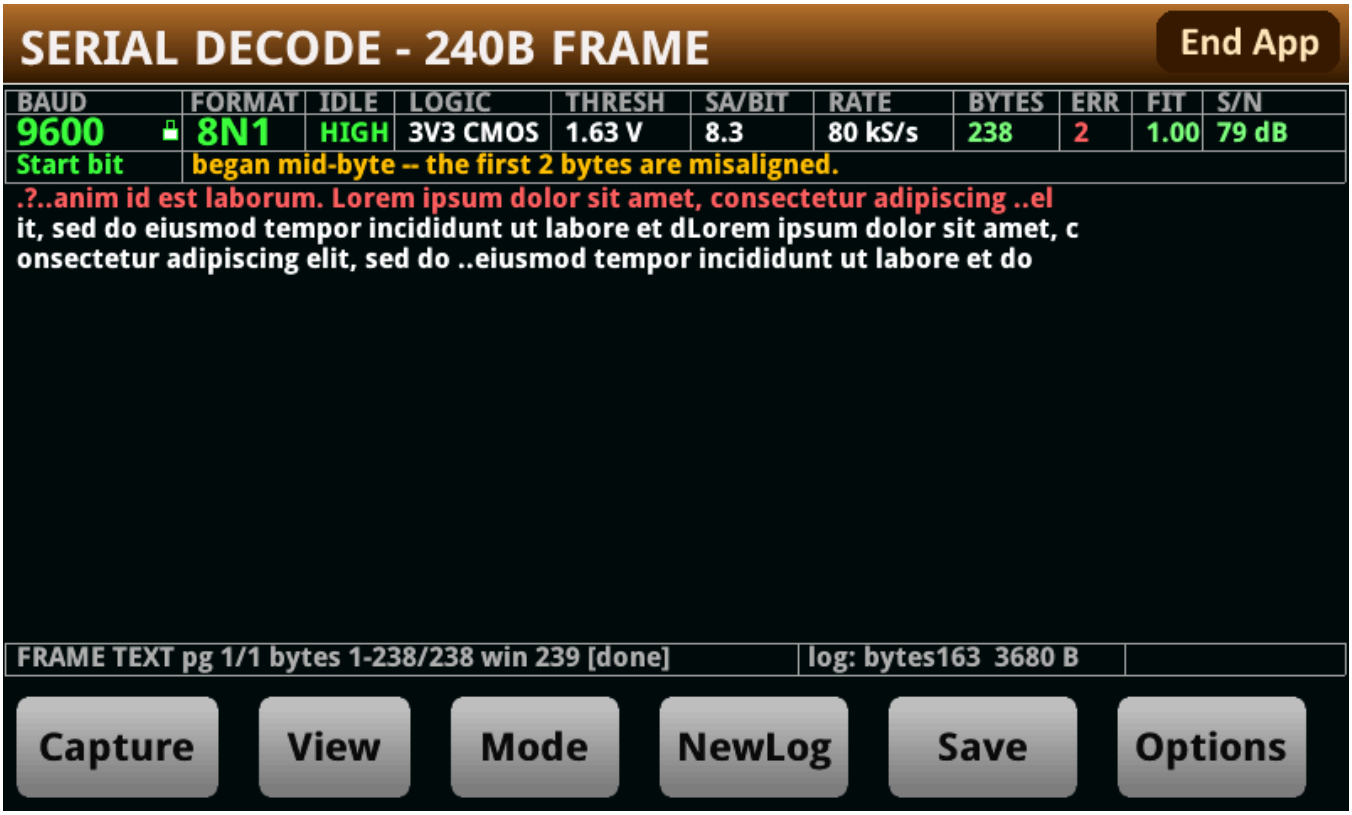
Which mode you are in is in the title bar — `SERIAL DECODE – 240B FRAME`, `– 8K CAPTURE` or `– 32K CAPTURE`, in the largest type on the screen.

The cell at the left of the note row is what a capture WAITS for, colour-coded so a glance is enough: green `Start bit` (the normal case — it waits for the line's own start bit), yellow `Free run` (it does not wait for anything), cyan `Trigger key` (it waits for the physical key).

The cell at the bottom right is the rear BNC, and it is **empty when the rear BNC is off** — which is the default. When it is not, it names the direction: green `TRIG IN`, amber `TRIG OUT`, cyan `TRIG I/O`. A trailing `?` means the input is switched on but *not in force*, which happens under `Free run` — free run means do not wait, so there is nothing for the input to gate.

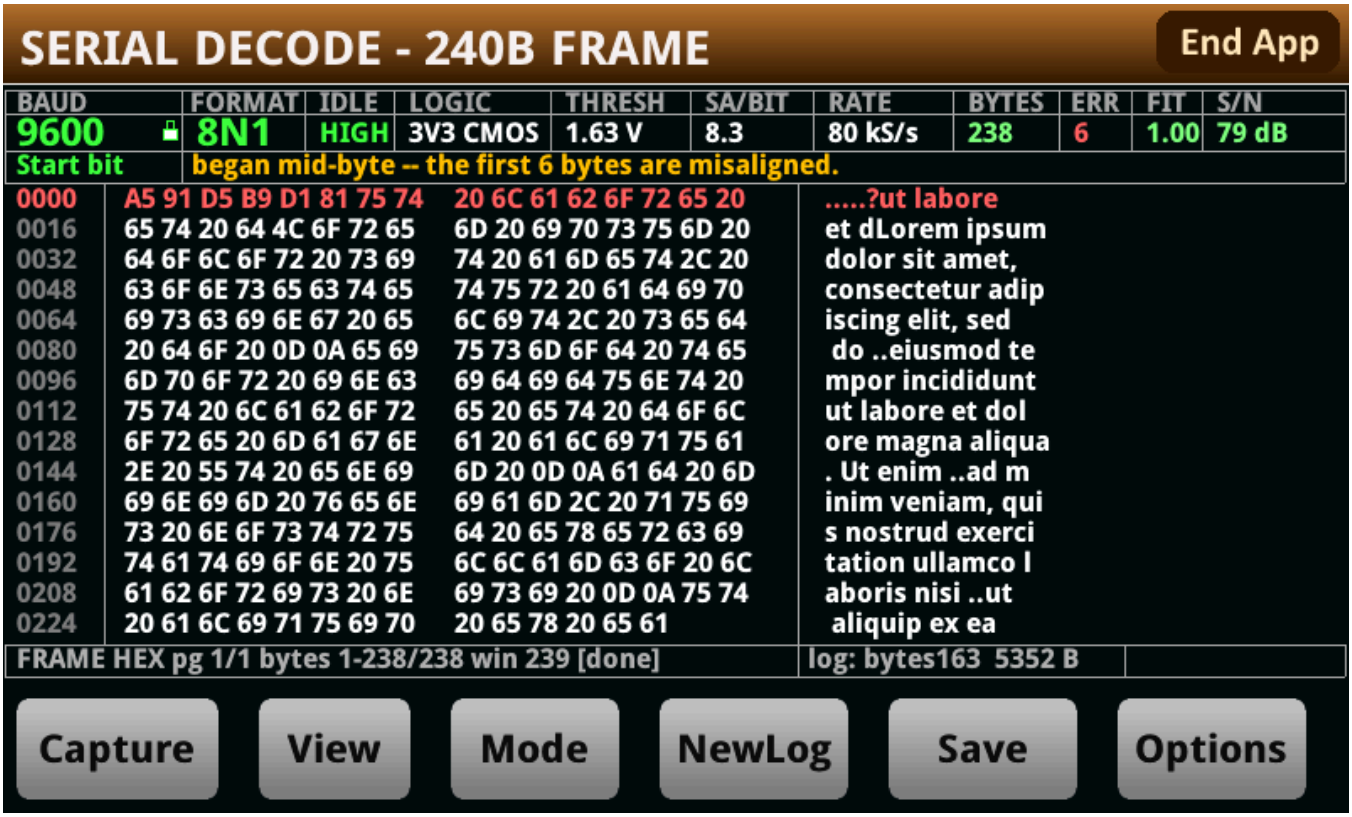
During a recording the status row carries a live counter — `recording... 40 % of the buffer`, then `decoding... 4096/8192 bytes` — with a **cyan progress bar** beside it, and the cell to its right reads `Capture or Mode = Stop`.

View switches the byte display to plain text, which is what you want for anything human-readable — and a row carrying a flagged byte is red here too:



The note row is the important one. Warnings appear there — an ambiguous baud rate, a line that disagrees with the rate you locked, a recording that stopped early. If more than one applies it ends with `(+2 more)`, and pressing **Save** writes all of them to a file.

Here it is doing that — a capture that started part way through a byte, which the app detected and said so rather than showing the misaligned bytes without comment — 13 of them, here:



An empty note row is not a promise that everything is perfect. It means the app did not detect anything wrong. A noise spike landing inside a data bit changes that byte with nothing to give it away.

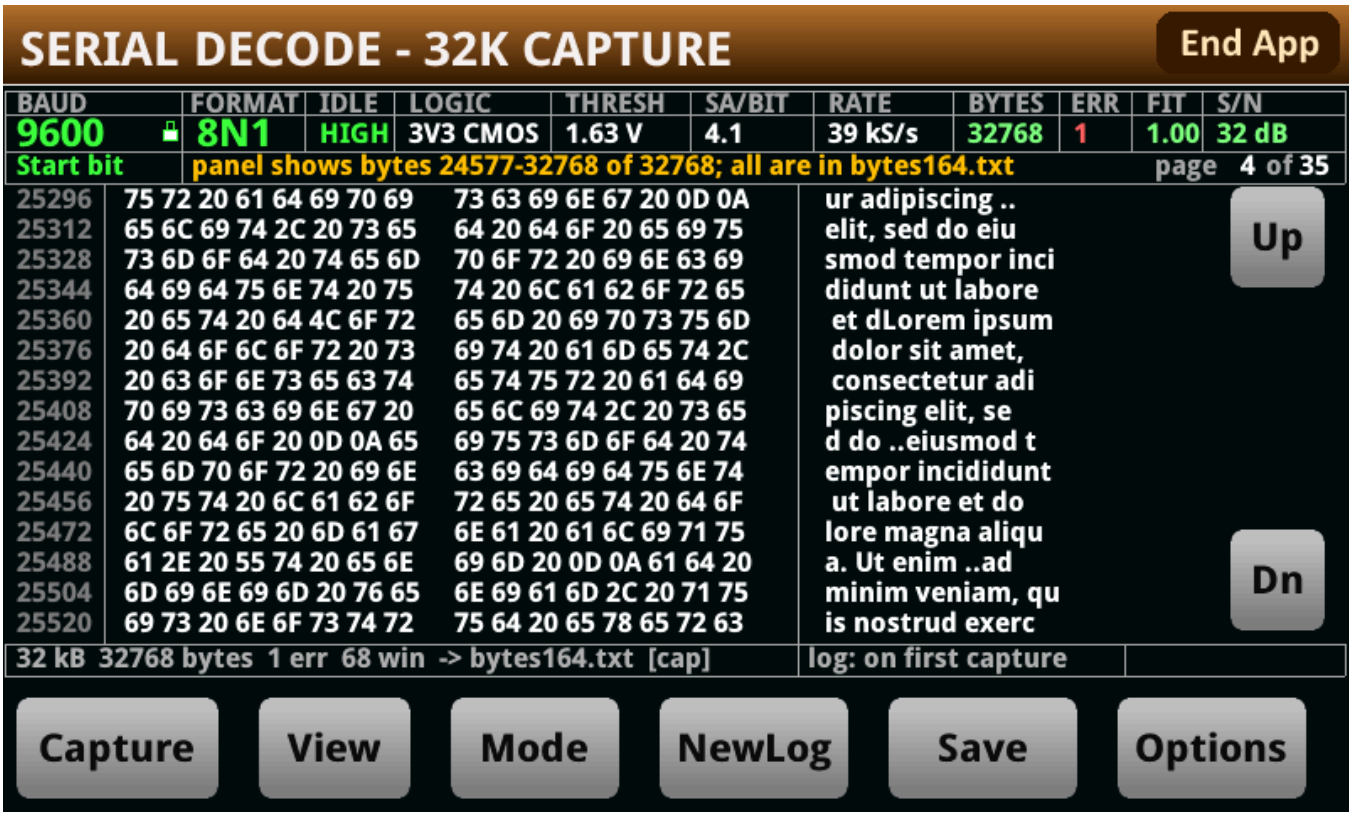
The buttons

Button	Does
Capture	Take a capture and decode it. In a recording mode, does the whole recording.
View	Switch the display: hex + ASCII → plain text → back.
Mode	Choose what Capture does: a screenful, an 8 kB recording, or a 32 kB one.
NewLog	Start a new numbered log file on the USB key.
Save	Write a full report of what is on screen to the USB key.
Options	The settings screen.
Up / **Dn**	Right-hand edge. Only appear when there is more than one page. Once you press either, **page N of M** appears at the right of the note row — and stays for later captures, which is why some figures here show it on a screen nobody has just paged.
Lock Rate	Right-hand edge. Only appears when the rate is not locked.

Every button responds immediately except the ones that take a capture. **Capture takes about 1.7 seconds when the rate is locked, and 4 to 9 seconds when it is not** — because to find an unknown rate it has to capture two or three times. Slower lines take longer, not less. The exact time varies from press to press because it depends on where the traffic happens to be when you press.

A press during a capture is not lost — it is queued and acted on when the capture finishes. So if the panel seems to ignore you, wait rather than pressing again.

Paging through a long capture



A recording is far more than one screen, so **Up** and **Dn** appear down the right-hand edge and **page N of M** appears at the right of the note row — the two numbers in white, the words dimmed, so the count reads at a glance. The note beside it says which bytes are on screen and that the whole run is in the file, which is the thing you cannot otherwise tell from a screenful taken out of the middle:

```
panel shows bytes 24577–32768 of 32768; all are in bytes054.txt
```

The offset column agrees with it — the rows start at 25296, not at zero — and the status row carries the run's own summary: 32768 bytes, 1 it cannot vouch for, 68 windows, and the file it went to. **That 1 is the same figure as** `ERR`, not a second opinion about it.

You page through the tail, not the whole run. The panel keeps the last **8 192 bytes** of a recording however long the run was; the file keeps all of it. That is why this 32 768-byte capture shows bytes 24577-32768 and pages 35, not 137 — and why the count is the same in either recording mode, since 8 kB is its own tail.

The page count follows the view. Press **View** on that same capture and it becomes 7 pages instead of 35, because a text row holds 80 characters and a hex row holds 16 bytes — 1200 bytes a page against 240, over the same 8 192:



Nothing about paging changes the capture; only how much of it fits on a screen, and the byte range in the note row is the same 24577-32768 either way. The two figures above are separate 32 kB recordings of the same line, so their `ERR` and filenames differ — the page counts are what to compare.

The TRIGGER key — the one hardware button that matters

The **TRIGGER** key is the physical key on the front panel, to the **right** of the screen — the lowest of that column of buttons. It is not part of the app's own row of buttons. It was intended to do two things nothing on the glass can do; **one of them works** — timing a capture — and the other, stopping a long job, does not. Both are described below, because knowing which is which is the difference between waiting confidently and pressing a key that does nothing.

1. It does not stop a long job, and nothing else does either

A recording runs to its own end. Once you press **Capture** in `8 kB` or `32 kB`, nothing on the instrument will cut it short — not a button on the glass, and not the TRIGGER key. Decide before you press. The status row says so before you commit:

```
32 kB  ready -- 32768 bytes to a file, no stop once started
```

There is deliberately no time estimate. Two different durations apply and only one is knowable here: the time your *device* takes to send that many bytes, and the meter's decode, which costs several times more — 8 kB is a couple of seconds on the wire and around half a minute to decode. A single number would be reassuring and wrong, so what you get is the exact byte ceiling and a progress bar.

What does bound a run, so it can never hold the panel indefinitely:

- the byte ceiling for the mode — 8192 or 32768;
- a quiet line: one second with no activity ends it;

- under flow control, 32 windows or 20 minutes of wall clock, whichever comes first. The 20 minutes is what matters on slow lines, where a full 32 kB window would take 18 minutes at 300 baud.

A tap on Capture during a run is not a stop. It is queued by the instrument and delivered when the run finishes. The app recognises presses made during the run it has just finished and discards them, saying so on the note row, rather than starting another recording you did not ask for.

2. It can time a capture

Set `Options ▸ Trigger = Trigger key`, press **Capture**, and the capture waits for you: it fires when you press the physical key, so you decide the moment the line is sampled. Verified on this hardware — armed on a deliberately idle line, the capture fired on the key press rather than timing out.

If the trigger you chose never arrives, the capture goes ahead free-running after a few seconds rather than failing, and the note row says which one was missing.

So the key has exactly one job here: it times a capture, and only when you have asked for that in `Options ▸ Trigger`. At every other moment — including all the way through a recording — it does nothing.

How much you can capture at once

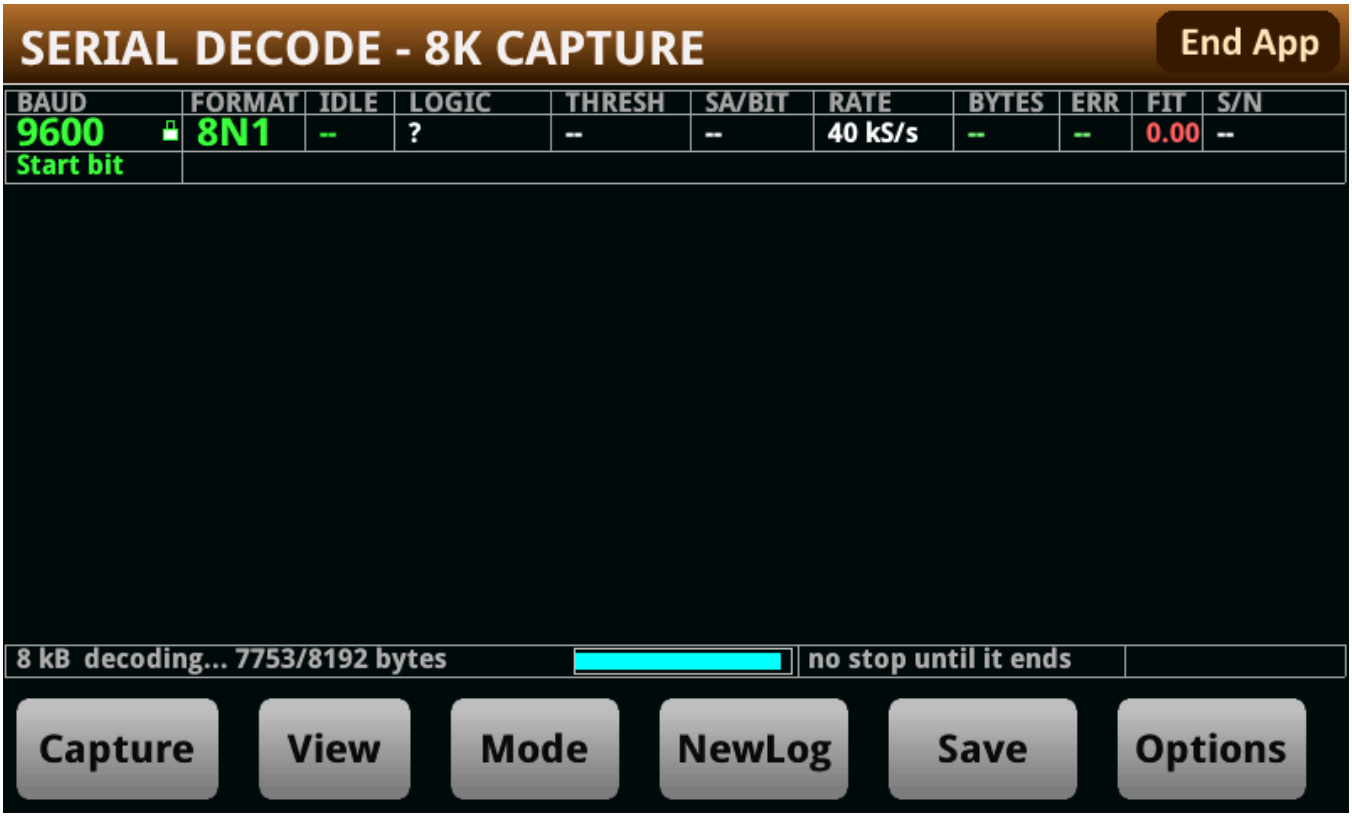
Four answers, and almost everyone only ever needs the first. **All of them are one press.**

You need	Use	Get
a look at what a device is saying	**FRAME** — just press Capture	~**240 bytes**, on screen
a longer burst, quickly	**Mode** → `8 kB`, then Capture	up to **8 192 bytes**, to a file
the longest single recording	**Mode** → `32 kB`, then Capture	up to **32 768 bytes**, to a file
more than that, losslessly	flow control, and a device that obeys it	as much as you like, ~20 minutes per press

FRAME is the normal case and covers almost all debugging. One press gives you a screenful, already decoded, with the baud rate and frame format worked out for you and a copy appended to the USB key automatically. You do not have to set anything up or lock anything down.

Recording to a file

Press **Mode** until the cell reads `8 kB` or `32 kB`. **You must lock the baud rate first** — press **Lock Rate**, or type the rate into **Options**. Then **press Capture once**, and that press does the whole job: it records, decodes every byte, writes them to the USB key, and comes back with the tail on screen. There is nothing to press in the middle.

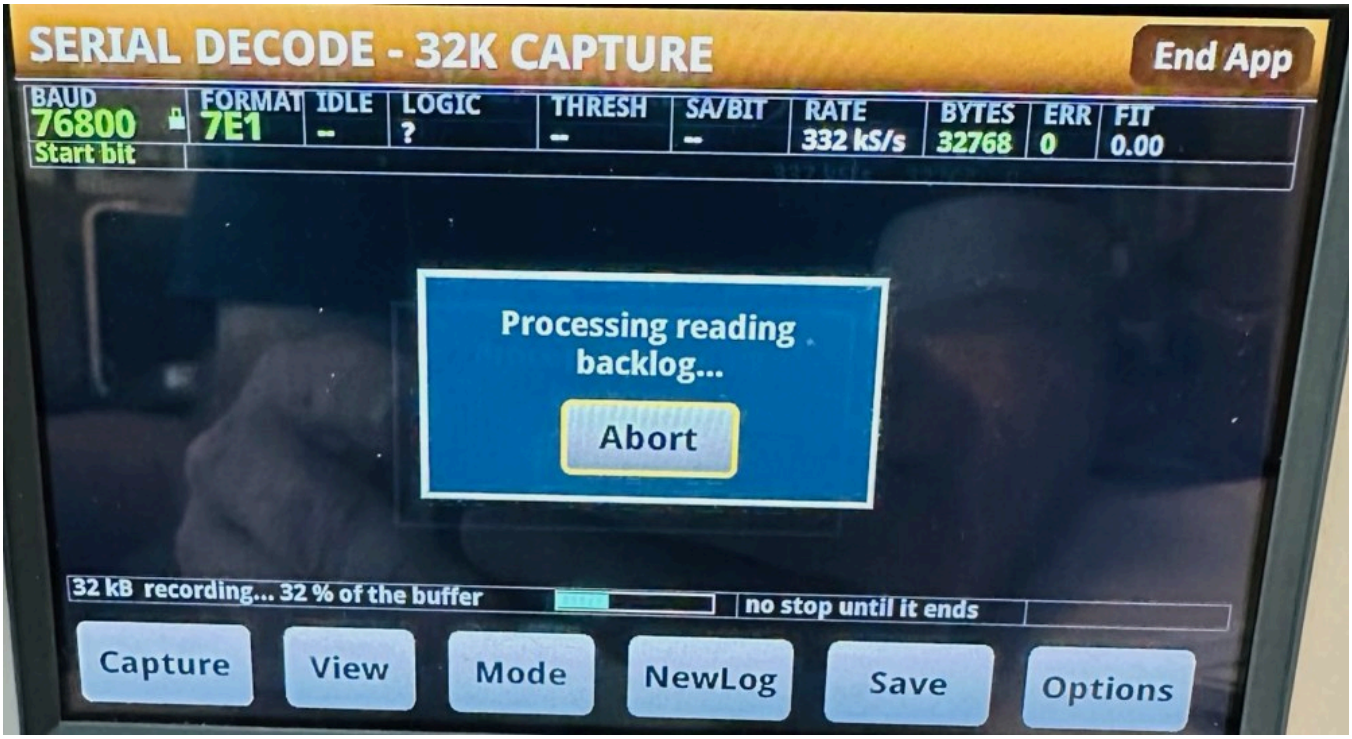


While it works, the counter on the status row moves — first `recording... N % of the buffer`, then `decoding... N/8192 bytes` — with a cyan progress bar beside it, and the cell to its right says whether anything can stop it. Above is a real 8 kB run in the *second* phase, decoding 5331 of its 8192 bytes, with `no stop until it ends` beside it. The measured cells read `--` and the dump area is blank because nothing has been decoded yet; the baud rate and format show only because they were locked rather than measured. **There is no way to finish it early** — see **The TRIGGER key** above for what bounds it.

When it finishes, the note row goes quiet. The status row's summary — bytes, errors, window count and filename — is the completion message; the yellow note row is reserved for what went wrong, so an empty one after a recording means there was nothing to say.

It stops by itself when the window is full or when the line has been quiet for a second. There is also a 20-minute backstop, so nothing can wait for ever — it only matters on very slow lines, where a full 32 kB window takes 18 minutes at 300 baud.

Nothing is dropped inside a recording. What you get is a continuous, in-order run of the line — verified with a 1024-byte non-repeating test pattern, where the decoded bytes were one unbroken slice of it.



> **If a `Processing reading backlog...` dialog appears, ignore it. Do not press Abort.**
>
> It is the instrument's own message, not the app's: the digitizer is producing readings faster than the meter is moving them out of the buffer. It **clears by itself** and the recording carries on normally.
>
> Seen on the bench from 38.4 kBd upwards — at 182 kS/s, at 635 kS/s recording a 76.8 kBd line, and within a second of pressing Capture at 1 MS/s on a 115.2 kBd line. Every one of those recordings completed and decoded correctly, including a full 32 kB one.
>
> Pressing **Abort** would stop the acquisition the dialog is describing, which is the one thing that actually loses the capture. There is nothing to do but let it run.
>
> The photograph above is that dialog over a live 32 kB recording of a 76.8 kBd line at 332 kS/s, with the progress bar behind it at 32 % of the buffer and the row beside it reading `no stop until it ends`. It is a photograph rather than a screenshot for a reason worth knowing: **the instrument's web interface stops answering while the digitizer runs**, so the remote screen grab cannot capture this dialog at all -- eight attempts timed out. If you need to document or report something that happens during an acquisition, a camera is the only way to see it.

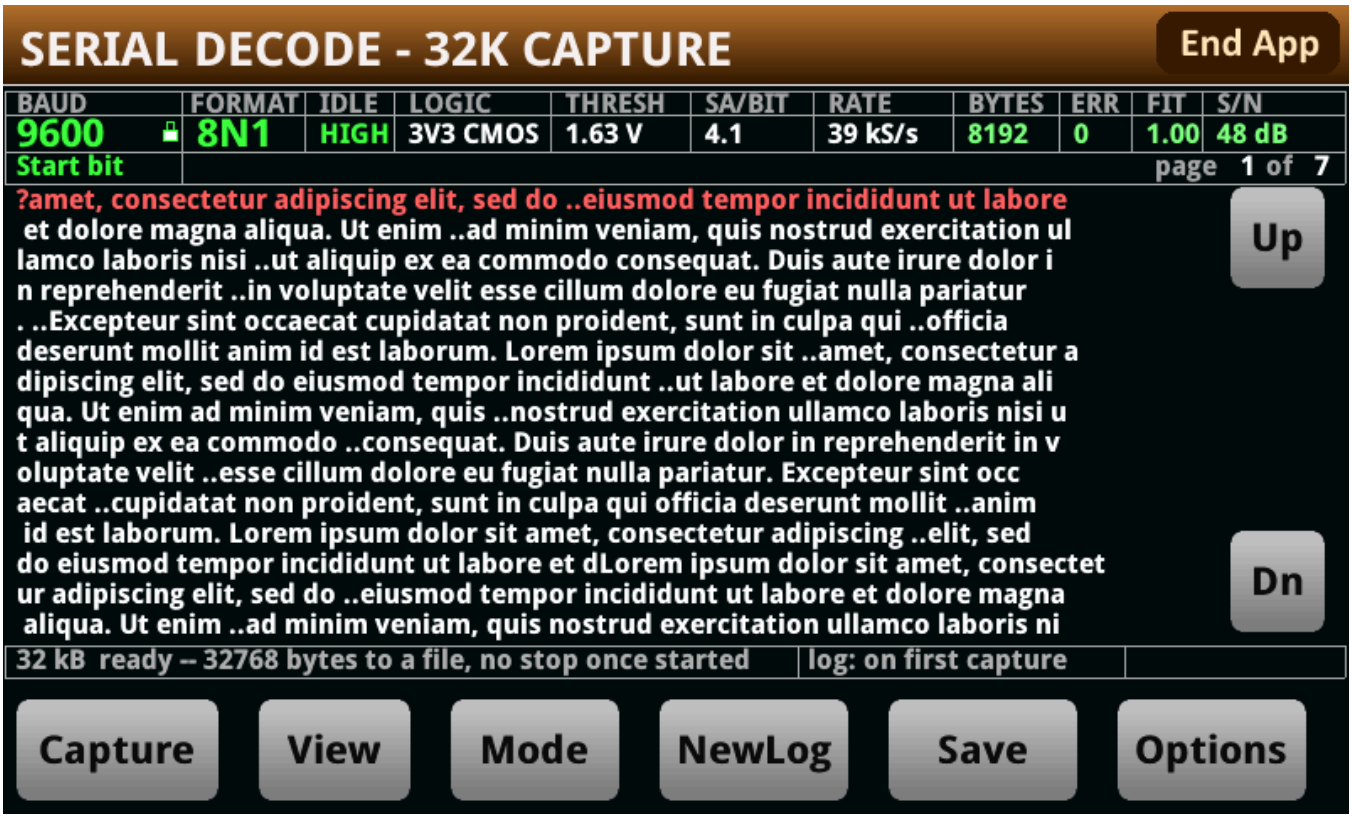
Which window size to pick

The window is the unit of work: one press records a window, decodes it, and files it. So the size is a straight trade, and neither choice affects what gets decoded — both are lossless.

	`8 kB`	`32 kB`
Bytes per press	8 192	32 768
Recording, at 9600 baud	~8.5 s	~34 s
Decoding afterwards	~30 s	~2 minutes
Best for	seeing something soon, and stopping soon	the most data for the fewest presses

Decoding is the slow part, and it is slower than the line: the meter decodes roughly 300 bytes a second whatever the baud rate. That is why `8 kB` feels much more responsive even though both are one press.

The app starts in FRAME, and one press of **Mode** reaches `8 kB`, a second `32 kB`.



Above: an 8 kB recording has finished and **Mode** has been pressed once, so the title bar reads `32K CAPTURE` while the previous run's retained tail is still on screen. `Up` and `Dn` have appeared because the tail is more than one page, the status row states the next run's ceiling and that nothing will stop it, and **the note row is empty**, because a recording that worked has nothing to warn about.

The mode you choose stays chosen. The title bar says which one you are in — `SERIAL DECODE – 240B FRAME`, `– 8K CAPTURE` or `– 32K CAPTURE` — and a recording that finishes leaves it alone, so pressing Capture again records another window. Only **Mode** changes it. Two things move it for you, and both say so on the note line: **Options ▸ Auto Detect**, because auto-detection cannot run in a recording mode at all, and clearing the baud rate, because a recording needs it locked.

More than one window: flow control

A window is not a total. If your device is built to wait, you can capture as much as you like without losing a byte, **and without pressing anything per window** — up to 32 windows or 20 minutes per press, whichever comes first. To carry on past that, press **Capture** again; it resumes where it stopped, in a new file.

Set `Options ▸ Rear BNC = FC Out`. The rear EXT TRIG OUT then emits **one pulse each time a capture is armed** (measured on a scope: idle-low at +0.20 V, +4.45 V high, 5.72 µs wide with a 436 ns rise — the width is the meter's own default and this firmware does not let the app change it) — a credit meaning "*send now, and send no more than one window's worth*". Then one press of **Capture** runs the whole conversation: credit, record, decode, file, credit again — round and round, with the window number on the status row, until the device stops sending. While the app is decoding, nothing is armed, no credit is issued, and a device that waits stays quiet, so the gap where bytes could be lost is closed by design. Everything lands in one file, in order.

It ends when your device goes quiet after a credit, or when it reaches one of two backstops: **32 windows** or **20 minutes**. There is no key that ends it early. Those exist so a device that never stops talking cannot leave the panel busy for ever, and the 20 minutes is usually the one you meet — decoding a window takes longer than recording it, so 20 minutes is about eight windows at 9600 baud rather than 32.

If it stops for either reason it says which, and how to carry on: **press Capture again.** A finished recording leaves you in the mode you chose, so it is one press, and the bytes that follow go into a **new** numbered file. **Nothing is lost in the gap** — your device only transmits when it is credited and no credit goes out until the next recording arms, so it is still waiting exactly where the run stopped. You get one conversation split across two files, in order.

Three things your device has to get right:

- **Catch a 5.7 µs pulse.** That is short: use an interrupt, an edge-triggered input or a hardware latch. A firmware polling a GPIO in a loop can miss it, and at 250000 baud 5.7 µs is under 1.5 bit times. It is a pulse per window, not a level you can poll once.
- **Send no more than one window per pulse** — 8 192 or 32 768 bytes depending on the mode. The pulse carries no size information, so the limit has to be built into your firmware.
- **Go quiet when it has nothing to send.** That silence is what tells the app the transmission is over.

Honest limitation: the pulse itself is verified — +4.45 V, 5.72 µs, 436 ns rise, one per arm, measured on a scope — and the loop that issues them is tested. The full round trip, with a device that actually waits for the pulse, has **not** been tested here: the signal generator on this bench stayed silent through a credit, and whether that is its trigger input or its burst arming has not been separated yet. The leading suspect is the width, since 5.72 µs is the narrowest pulse this bench can present and the generator's own minimum is not specified anywhere. So the mechanism is sound by construction and you would be the first to close the loop. **Triggering, in both directions** in REFERENCE.md has it gate by gate.

How long a recording gets you

Two different times, worth keeping apart: the **signal** that ends up in the file, and the **waiting** before the buffer is full. Below about 19200 baud they are the same. Above it they are not, because the meter writes about **100 000 readings a second** into its memory however fast the digitizer is told to sample. The samples are properly spaced and the record is complete — gathering them just takes longer than the stretch of line they represent.

Do not confuse that ceiling with the sample rate. The **sample rate** decides how many samples land in each bit, and it is what limits recording to 153600 baud (below). The **100 000 readings a second** is not about fidelity at all: it is how fast readings reach memory, and it decides only how long you wait.

Locked rate	Signal you get	Waiting for the full 32 kB
300 Bd	1092 s	the same
1200 Bd	273 s	the same
2400 Bd	136 s	the same
4800 Bd	68 s	the same
9600 Bd	34 s	the same
19200 Bd	17 s	the same
38400 Bd	8.5 s	about 17 s
57600 Bd	5.7 s	about 17 s

Locked rate	Signal you get	Waiting for the full 32 kB
115200 Bd	2.8 s	about 18 s
153600 Bd	2.1 s	about 21 s

Every row is the same 32 768 bytes — what changes is how long that takes to go past. The `8 kB` window is a quarter of each figure. Then add the decoding, which does not depend on the baud rate: about two minutes for 32 kB, half a minute for 8 kB.

You do have to fill it. There is no way to stop a recording early — see the TRIGGER key section above — so choose the window with the whole wait in mind.

153600 baud is the fastest rate that can record. Above that, FRAME only.

The reason is that a recording and a frame capture do not get the same sample rate out of the same request. A frame capture gets very nearly the 1 MS/s it asks for; a recording's burst acquisition loses about half a microsecond per sample, so the same request delivers about 662 kS/s. The floor is four samples per bit, and the crossover works out at **165563 baud** — so of the standard rates, 153600 is the last one that clears it, at 4.3 samples/bit.

Above it the app **refuses before you press**, on the note row, naming the figure a recording would actually get:

> `192000 Bd: a recording gets 3.4 samples/bit, under the 4 floor -- use FRAME`

That figure is deliberately the recording's, not the header's. The `SA/BIT` cell will read about 5.2 at 192000 baud, which is true of the frame capture it was measured from — the two are different acquisitions, and the note says which one it means.

FRAME keeps working to the full 250000 baud, and usefully so: at 192000 baud a frame capture holds 384 bytes.

Reading the bytes

The hex view shows two groups of eight bytes and then the same bytes as text. The offset down the left is dimmed because it is there to navigate by, not to read. A byte that could not be recovered shows as `--`.

`ERR 0` does not mean the bytes are definitely right. It counts broken stop bits and parity failures. Noise landing inside a data bit flips it with nothing to detect — which is why the note row matters more than the error count.

If you see **`began mid-byte — the first N bytes are misaligned.`**: on a line that never goes quiet, the capture starts wherever it starts, and if that is halfway through a byte the first few bytes are chopped up wrongly. They are shown rather than hidden, but **do not trust their values**. Everything after the first gap in the traffic is fine. On traffic with normal gaps between messages it costs **5 to 12 bytes**; on a line that never pauses it can be far more — **up to 165 bytes measured** on back-to-back text at 250000 baud, because there is no gap to resynchronise on. A device that pauses between messages never shows it at all.

Locking the baud rate, and the amber padlock

The app locks the rate for you after a good capture, which is why later captures are quicker. It only does this when it is confident: the rate landed on a standard one, enough clean frames came through, `FIT` was 0.9 or better, and the voltage levels were steady.

When it is not confident it leaves the padlock **amber** and offers you the **Lock Rate** button. The bytes are still shown — they just are not trusted enough to carry forward. Press **Lock Rate** to pin it anyway, or capture again.

Auto-lock never locks polarity. It is worked out fresh on every capture, on purpose: guessing it wrong once and then sticking to it is how you get a confident, completely wrong decode. (**Lock Detected** on the Options screen *does* copy the detected polarity into the fields along with everything else — but there you can see what it picked before you commit it.)

If the line changes rate, the app gives the lock up

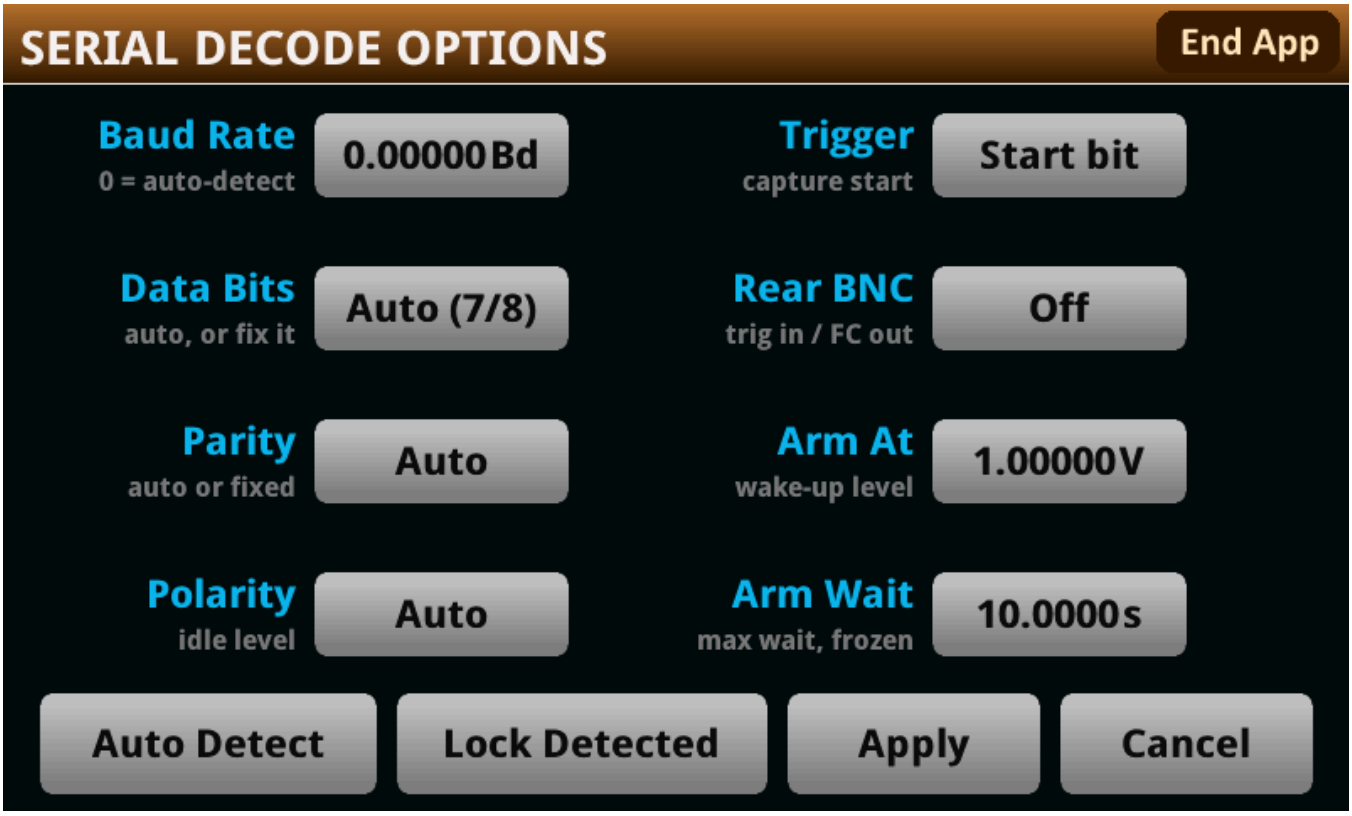
A locked rate is a promise about the wire, and when the wire stops keeping it the lock is worse than useless — it turns every frame into an error and nothing gets better by pressing Capture again. So when **all three** of these hold, the app **drops the lock by itself** and re-detects from the same samples:

- a rate is locked;
- the bit time it measures on the wire disagrees with that rate;
- more than a quarter of the frames in the middle of the capture failed.

It says so on the note row, naming the rate it discarded — `19200 baud fit nothing -- unlocked, and this capture reads 38400` — and from there auto-lock pins the new rate on the usual terms. That is the same result as typing `0` into Options by hand, which is what you had to do before.

The middle condition is what protects a rate you typed deliberately. If the rate fits the wire and the *format* is wrong, every frame fails too — but the measurement agrees with your number, so your number is kept.

Options



Setting	What it does
Baud Rate	`0` means auto-detect. Type a number to lock that rate; it is rounded to a whole baud.
Data Bits	Leave on `Auto (7/8)`. Only use `Auto (any)` if you know the device sends 9-bit words — it can make a damaged capture look tidier while being wrong.
Parity	`Auto`, or pin `None` / `Even` / `Odd`.
Polarity	`Auto`, `Idle high` (normal CMOS/TTL) or `Idle low` (inverted). Auto is reliable unless the line is almost never idle.

Each setting is independent — pinning one leaves the rest on auto.

Apply commits your changes and takes a capture. **Cancel** puts everything back to how it was when you opened the screen. **Auto Detect** forgets everything locked, takes a fresh capture, and locks whatever that finds — it is the quick way back when you have locked onto the wrong thing. **Lock Detected** copies the last capture's settings into the fields so you can look at them before committing.

Apply takes as long as a capture, because it is one. Measured on the bench: about **2 seconds** with the rate locked, and about **4 seconds** with nothing locked, because an auto-detecting capture costs a whole extra acquisition to find the rate plus a format search. The screen returns to the main panel immediately and the status row reads `capturing` for the rest of it. If Apply feels slow, lock the rate — it is the same saving every capture makes, a bit over two to one here. Nothing else on this screen takes a measurable amount of time: moving between the fields runs no app code at all.

There is no stop-bit setting. A second stop bit looks exactly like a bit of idle line, which the decoder already copes with, so a 2-stop device decodes correctly and is reported as 1 stop.

Advanced

Four settings you can ignore unless you need them:

- **Trigger** — what starts the acquisition. `Start bit` (the default) waits for the line to move. `Free run` grabs immediately. `Trigger key` waits for the front-panel TRIGGER key, as described under **The TRIGGER key** above. **If the thing you asked for never happens, the capture goes ahead anyway** — after a few seconds, or after **Arm Wait** on a line that has not started — and the note row names the one that did not arrive: `edge trigger unavailable; captured free-running`, or `front` for the TRIGGER key. Those bytes are real but they are not lined up with the event you wanted.
- **Rear BNC** — lets an external trigger in on the rear connector, and `FC Out` sends the flow-control credit pulse out of EXT TRIG OUT. `FC Out` is what turns a recording into an unlimited one; see **More than one window: flow control** above.
- **Arm At** — how far from the idle level the capture waits for the line to move, in volts, from 0.33 V to 6 V. Default 1 V. Only used on a line that is **not transmitting yet**; see **Waiting for a device that has not started** below.
- **Arm Wait** — how long such a capture may wait, from 2 s to 120 s. Default 10 s. **The panel is frozen for the whole of it**, which is what the field's help text means by *max wait, frozen*: at the 120 s ceiling that is a two-minute dead panel. It lengthens the wait and never shortens it, so an edge capture always waits at least its own 3 s — 2 s and 3 s do the same thing, and the note after an expiry reports the 3 s it really waited. It governs `Start bit`; `Trigger key` ignores it and waits its own 30 s for a finger.

Waiting for a device that has not started

Press Capture before the device powers up and the capture waits for it instead of recording the silence. It is the ordinary case on a bench: a board that talks when you switch it on, or when some event happens to it, and nothing on the line until then. It needs **Trigger** on `Start bit`, which is the default: `Free run` grabs immediately by definition, and `Trigger key` waits for your finger instead of for the device.

What the app does with it:

- The level probe finds no transitions, which is how it knows. On a line that is already transmitting nothing here applies — the threshold is measured from the two levels, which beats any fixed voltage.
- The comparator is armed **Arm At** volts from the idle level, on the side the start bit travels: one volt above a line sitting at ground, one volt below an idle-high TTL line, one volt above an RS-232 line idling at –6 V. On a line idling away from ground the level is clamped at the midpoint, because the far level is unknown until the device speaks.
- The window opens **before** the first start bit, not on it, so the first byte is whole. The trigger model keeps 5 % of the buffer back as a pre-trigger reserve — on a busy line almost none of it is spent, and on a silent one all of it is: **1055 samples** of quiet line ahead of the device's first edge.
- The panel says it is waiting: the right-hand cell reads `waiting 10s for 2.30V -- TRIGGER=go` for the length of the wait. **The panel is frozen while it waits** — a touch press only queues — so that line is the only thing that distinguishes a live arm from a hung instrument.
- **The front-panel TRIGGER key cuts the wait short**, in every setting this screen can produce. It is the one escape that works: the key is a stimulus of the instrument's trigger model, serviced in firmware, so it fires while the app is blocked. Pressing it captures whatever is on the line at that moment.
- **Where in the wait the device starts does not matter.** Against a 10 s wait the arm fires at +0.3 s and still fires at **+9.7 s**; at +11 s it expires.

- If **Arm Wait** runs out **the capture still happens, free-running**, and the note row says the whole of it: `edge trigger unavailable; captured free-running (no trigger in 10 s (edge) -- nothing crossed 0.99 V; raise Arm Wait or lower Arm At)`. That names both settings and the level the comparator was really holding, which is the pair you can do something about.

The rate is a guess on this path, and the panel says so. Nothing can measure the bit rate of a line that has not started, so the capture runs at 200 kS/s — 20 samples a bit at 9600, and enough for anything up to 38 400 — and the note row reads `armed at 200 kS/s on a line that has not started -- lock the baud rate for a longer window`. That advice is the useful one: **lock the rate in Options and the capture uses the right one for it**, with a full window and no guessing. Locked, it is reliable right across the range — **0 bad frames** at 1200, 9600, 38400 and 115 200 baud.

If the bit rate cannot be measured from what arrives, the app tries 9600, 19200, 28800, 31250 and 38400 against the samples it already has, and publishes one only when exactly one of them frames the bytes cleanly — `decoded at an ASSUMED 9600 Bd; lock the rate to be sure`. Where two rates fit equally it says so and decodes neither, because a capture that fits 19200 and 9600 alike cannot be told apart from inside the instrument. **Past the top of that ladder it refuses rather than guess**: a 115 200 Bd device gives `1.7 samples/bit -- 115004 baud needs a faster capture than 200 kSa/s` — the second reason to lock a rate you already know.

Recordings wait too. An 8 kB or 32 kB recording started on a silent line arms the same way, records from the device's first byte and decodes it cleanly. While it waits, the recording's own limits are suspended — the idle watchdog cannot end a run before the device has started — and an arm that expires ends the run naming the same two settings: `nothing crossed 2.30 V in 10 s -- raise Arm Wait or lower Arm At`.

An external pulse on the rear connector really does start a capture — verified on this hardware, and verified by the negative as well: take the pulse away and the capture refuses rather than quietly grabbing something, with the note row naming what was missing. Two things to know before you rely on it. The rear input is **OR'd with** whatever **Trigger** says rather than replacing it, so the first of the two to happen wins the capture — and on a line that is always transmitting that is the start bit, every time, which leaves your pulse no say in where the window opens. Point it at a line that is otherwise quiet, or pair it with `Trigger key` so that the only thing it competes with is your own finger. And under `Free run` the rear input is ignored altogether, because free run means do not wait for anything — the app says so rather than leaving you to notice: the note row reads `Rear BNC = Trig In is set but IGNORED`, and the status cell marks it by appending a `?` to `TRIG IN`.

Trigger applies to FRAME captures, and to a recording in one case: a line that is **not transmitting when you press Capture**, where the recording waits for the device exactly as a frame capture does. A recording of a line that is already talking ignores it and starts immediately — there is nothing to wait for, and for a flow-controlled recording the thing that decides when the device talks is the credit pulse, not a trigger. `FC Out` is not confined that way — it pulses once per armed capture whichever mode you are in, a FRAME capture included. What a recording adds is the repetition: credit, record, decode, credit again.

Files on the USB key

Two different files, and they are not the same thing:

File	Written	Holds
`/usb1/SERDEC/bytesNNN.txt`	automatically, on every capture	the running log — every capture's bytes as text and hex. **NewLog** starts a new number.
`/usb1/SERDEC/serial_NNN.txt`	only when you press **Save**	a full report of what is on screen: settings, all the notes, the whole dump.

Everything goes in `/usb1/SERDEC`, and the app creates that directory when it starts — there is nothing to set up, and nothing lands in the root of your key. The panel still shows the bare filename rather than the path, because the directory never changes and the row is narrow.

You can take the key out and put it back while the app runs

Pull the key whenever you like. Within half a second the `Save` and `NewLog` buttons disappear, the log cell reads `log: no USB key`, and the app stops writing. Put a key back and the buttons return, a fresh `bytesNNN.txt` is started on it, and logging carries on. Nothing needs ending and restarting.

The app checks the slot twice a second while it is sitting idle, which is the only way to notice: the instrument posts no event when a key arrives or leaves, so there is nothing to listen to. The check costs about 340 microseconds, or 0.07 % of the instrument. It stops while a capture or a decode is running, and resumes when that finishes — so a key pulled *during* a long recording is noticed by the write that follows it rather than by the clock.

A bad install bluescreens the instrument rather than refusing. Two versions of this have been seen on firmware 1.7.17a, and the second matters more because it looks innocent:

- **A corrupted archive on the key.** Recopy `Serial_Decode.tspa` and verify it with `cmp` before installing again — see the note in the README's install steps.
- **A capture taken while the app starts up, run from internal memory.** This one is fixed in the app, and it is worth knowing about because the fix is a single missing call rather than anything you can configure. Versions up to 1.37 took a capture at the end of their own launch. Installed in internal memory, that bluescreened the instrument at the end of that capture — three times out of three — while the *same archive* run from the USB key was faultless, and a capture from your own press of **Capture** was faultless too. The app no longer captures at launch, which removes it: the identical build with only that call taken out ran clean.

  Two things about it are worth recording, because both cost a day to establish and both are counter-intuitive:

  - **The installed copy was not corrupt.** Its stored source was byte-identical to the working one — 271345 bytes, 7164 lines, the same CRLF endings, not one differing byte.
  - **It is not a size limit, so making the app smaller does not help.** A purpose-built app with a *larger* compiled image and *more* stored source runs cleanly, and stripping 58 kB of comments out of this app changed nothing while the launch-time capture was still there.

  Reinstalling from the key is **not** a reliable cure for this — one reinstall looked like one and the crash came back, so never conclude it is fixed because a capture or two survived.

So `tools/cmp_installed.py` proves the TEXT, not the INSTALL. It reads the installed app's own source back out of the script catalog and diffs it against any commit's archive down to the byte, which settles exactly which build is on the instrument. It cannot see whatever else the firmware keeps beside that text, and that is what a bad install damages. **Reinstall from the key before suspecting the code** — it costs a minute, and it is the remedy that has worked both times.

Expect about two seconds before the buttons come back after you push a key in. That is the instrument mounting the key, not the app waiting — pulling the key is noticed at once, and the delay is only on the way in. Pressing a button during those two seconds does not shorten it: every part of the app, the buttons and the clock alike, has to ask the same question of the firmware, and until the firmware says a key is there the answer is the same whoever asks.

Tidying the key does not help, so do not bother on this account. Measured over seven pull-and-replace cycles with the key's root cut from 30 entries to 16 — 400 files deleted and both firmware `.upg` images removed — the delay was unchanged. Across those cycles the instrument reported the key *absent* about four times as long as *present*, on a rhythm that was close to even by hand: it is the mount, not the amount of data to be scanned.

`Capture`, `View` and `Mode` also check the slot themselves, before doing anything else, so a press never waits on the clock. This is belt and braces rather than a fix for something you would have seen: a capture already re-opened the log and brought the buttons back before it started work. What it changes is that each press asks, instead of getting the right answer as a side effect.

Every write is checked first, so a pull does not produce a pile of pop-ups. The instrument's `file.write` does not report a failed write to the program at all — it posts `2200, File write error` on the panel instead, once per call — so the app asks whether the key is there before each one. Pull the key mid-recording and the run stops writing and reports the file as incomplete; before that check existed, a 32 kB recording could put a modal dialog on the panel for every one of its 2048 rows. Anything still in flight at the instant the key leaves may post a single one.

The key that comes back is treated as a different key. The app forgets the filename it chose on the old one and allocates a new one, because the old name may already be in use on the new key and appending to a stranger's file is worse than starting another.

With no key in the slot, `Save` and `NewLog` are hidden. Both exist only to put a file on the key — Save writes the report, NewLog starts the next byte log — so with nowhere to write, neither has a press that could succeed. Capture and decode work perfectly well without a key, in **every mode**; you simply get no log and no report. A 240-byte capture behaves exactly as usual, and the 8 kB and 32 kB recordings still acquire and still decode — the bytes go to the panel instead of to a file, and the status row says `NOT recorded` rather than naming one. The panel keeps the last 8192 bytes, so an 8 kB recording is kept whole and a 32 kB one shows its last quarter; the note row says which bytes you are looking at.

You may also see `2205, File not found` during normal use, with a key inserted. Choosing the next `bytesNNN.txt` means asking the instrument about names until it finds one that is free, and the instrument reports each name that is not there. It is a question, not a fault.

With a key in, a recording decodes into the log file, so the file holds the whole run while the panel shows only the tail of it. The note row says so when they differ. Without a key the tail is all there is, and the note row says that instead.

A recording does not need a key — it is better with one, because the file is the only place the whole run exists. Every mode captures and decodes without one; what you lose is the log and Save.

When something looks wrong

What you see	What it means
`line is idle (no transitions)`	Nothing is crossing the threshold. Check your connection, and check the device is actually talking while you capture.
`no clear logic levels`	Too much noise, or the swing is too small. 0.33 V works on the bench, and 0.25 V did too; 60 mV is refused outright.
Wrong rate reported	The traffic may be genuinely ambiguous (below). Lock the rate if you know it.
Bytes wrong but `ERR 0`, with the rate locked	The device is not running at the rate you locked. The note row warns when the line disagrees by more than 4 %.
A baud rate that is not a round number	It is reporting what it measured, because the measurement was not within 2 % of a standard rate.

Some traffic is genuinely ambiguous, and no decoder can fix it. Eight `0x00` bytes at 9600 baud 7N1 make exactly the same waveform as four `0x08` bytes at 4800 baud 8N1. The app tells you a rival rate fits rather than pretending to know. Irregular gaps between bytes remove the ambiguity completely.

Three cases where automatic detection is known to fail are described under **Known failures of automatic rate detection**, below. All three are fixed by typing the rate in.

What it can cope with

Measured at 8 samples per bit. Past every one of these limits the app **reports** the problem rather than quietly getting it wrong, except where noted.

Condition	Works up to
Wrong clock, rate **locked**	**±4 %**. Beyond that a fast device corrupts bytes with nothing to show it
Wrong clock, **auto-detect**	±12 % and past — it measures instead of assuming
Jitter	**±15 %** of a bit; past ±20 % it can fail quietly
Noise	about **40 %** of the logic swing — roughly 1.3 V on a 3.3 V line
Slow edges, long cables, weak pull-ups	filtering up to **0.7 bit times**
Logic swing	**0.33 V to 8 V** — the limit is the swing, not the logic family
DC offset, logic low at or above 0 V	anything that keeps the line inside ±10 V
DC offset, logic low **below** 0 V	works, but see the note below — one band is not reliable

Rates from 300 to 250000 baud decode byte-exact on the bench, and so do odd ones — 900, 1500, 3600, 7200 and arbitrary values like 1379, 8123 and 104857. A 1 kB non-repeating payload is clean at every one of the eleven standard rates from 300 to 250000, and 30 consecutive captures at 2400 baud returned ~6 780 bytes with nothing flagged. A rate within 2 % of a standard one is reported as the standard one, because real devices use standard rates.

Tolerance follows samples per bit, not baud rate. A slow line is not automatically the safe one — read `SA/BIT`.

If you are writing the firmware at the other end, two things help: keep your edges within ±15 % of a bit time (that is ±15.6 µs at 9600 baud but only ±1.3 µs at 115200, so if an interrupt can delay your transmit loop, send slower), and avoid padding with `0xFF`, which gives the rate detector only one narrow pulse per frame to measure. `0x00` padding is fine.

Known failures of automatic rate detection

Read this in proportion. Most serial decoders require you to tell them the baud rate. This one measures it, and gets it right on every standard rate and on arbitrary ones like 1379, 8123 and 104857. Three cases are known where the measurement goes wrong — **8 points in 860** on a full bench plan for the first two, and 3 in a million for the third — and none of them is an ordinary payload at an ordinary rate. In all three the remedy is to supply the rate, which is what other tools ask for as a matter of course.

All three are fixed by typing the rate into `Options ▸ Baud Rate` — verified, every frame decoded with zero errors once the rate was given. The numbers are below so you can judge the risk yourself.

Automatic detection has no clock to refer to. It works in two steps: it measures the gaps between transitions and finds the longest bit time that divides them all, then weighs that answer against the standard rates, because a real device almost always uses one.

Gaps of many different lengths pin the measurement down well; gaps of only a few lengths do not. But where detection fails it is usually not the measurement that is wrong — it is the second step. A different reading at a simple multiple of the measured rate can land on a standard value while the true rate does not, and that coincidence is allowed to outweigh a correct measurement.

Two checks stand between a correct measurement and its own multiples, and both apply only once the measurement itself lands on a standard rate — where the true rate is non-standard there is nothing to defend, which is why the failures below concentrate there.

- **Where the pulses fall.** At the true bit time some pulses span an odd number of bit cells; divide that bit time by an even number and every pulse spans an even count. So a halved, quartered or sixthed reading is recognisable from the pulse lengths alone.
- **How short the shortest pulse is.** Dividing by an *odd* number leaves the counts' parity exactly as it was, so thirds and fifths are invisible to the first check. What gives them away is that they claim a longer minimum. Ordinary traffic has single-bit runs — a ten-bit frame opens with a one-bit start bit — so a reading whose shortest run is three or five bit times is describing something else, and a reading past about two and a half is refused. **The cost is real and worth knowing:** a payload whose shortest genuine run *is* three bit times or more, such as `0x00` sent with a two-bit gap between bytes, cannot have a third-of-the-rate reading accepted. Such a payload reports a third of its true rate either way, so this refuses nothing that was working, but it is the shape to watch for.

1. A short pattern repeated over and over

What triggers it. Data built from `0x00`, `0xFF`, `0x55`, `0xAA`, walking bits, or one fixed string sent on a loop — link-training patterns and idle fill, rather than real messages. Such data yields only a handful of distinct gap lengths, and a simple fraction of the true bit time then explains those gaps just as well as the truth: half the bit time reads the same square wave as twice as many bits, with no framing error anywhere to give it away. Both readings describe the waveform, so nothing in the signal can rank them — and where one of them lands on a standard rate and the true rate does not, the standard one wins.

The measurement is not the problem. Traced on a failing case, the bit time was recovered as 8.3990 samples against a true 8.3987, every one of the smallest gaps sat at the true bit time, and the decode at that rate framed every byte without a single error. It was then overruled by its own half, because 3572 is not a standard rate and 7200 is.

What you see. The reported rate is a small whole-number multiple of the true one — two to five times — and always one of the standard rates. Expected 3572 and the panel says 7200, or expected 124000 and it says 250000. Usually the byte area also fills with error markers, because a wrong bit time shreds most frames; in the measured cases 83 to 119 frames of a few hundred were flagged. Occasionally the frames survive and the bytes are simply wrong, which is the case worth knowing about.

There is no warning about the rate itself. Where two rates fit a waveform the panel normally names the runner-up under the reading, but not in this case — verified, the note is empty here. So the error markers are the whole of the signal, and where the frames happen to survive there is none. If a repeating test pattern reports a standard rate you did not configure, distrust it and lock the rate.

How rare, and where. Measured over **1 677 waveform-and-rate cells, each captured from 200 different points in the waveform — 335 400 captures.** The capture's opening point matters more than anything else here, because it decides how much of the first byte is missing and therefore how damaged the reading looks:

the true rate is	captures	reported wrong	
a **standard** baud rate	171 600	882	**0.51 %**
a **non-standard** rate	163 800	4 554	2.78 %

Every one of the 882 is an impaired signal, not merely a repeating pattern: three waveforms carrying 0.6 V of sinusoidal drift and one carrying heavy edge jitter. **No ordinary payload misreports at a standard rate at all** — the fox, the 1 kB text payload, twelve random payloads and the walking-bit patterns are clean at every standard rate from 300 to 250 000 baud, across all 200 opening points. That is 171 600 captures, and the split above is drawn from the same run.

A **LIN-style break field** — 13 or more bits held low, which is not valid 8N1 at all — is the hardest case the bench carries, and it reads correctly at 600, 19 200, 31 250, 38 400, 57 600 and 76 800 baud from every one of the 200 openings. This decoder does not claim to read LIN, and a break field is outside UART by construction; if you are probing a bus that begins frames with a long dominant field, the rate is worth setting by hand.

On the instrument the same plan gives 8 wrong rates in 860 capture-and-decode points, on three waveforms: the repeating single-value blocks, a LIN frame, and a heavy-jitter case.

One qualification, which is not this failure. Half of a standard rate is often *also* standard — 4800 is half of 9600 — and for perfectly regular traffic the two readings are the same waveform, as **Some traffic is genuinely ambiguous** above describes. That is not a misdetection and the panel names the rival rate when it happens; the failure in this section is the one where it does not.

The bench deliberately spends **half** its points on non-standard rates, which is far more than ordinary use, so the figure above overstates the exposure of a typical setup by roughly five times.

Besides locking the rate: capture real traffic rather than a test pattern, or let the bytes have irregular gaps — either removes the ambiguity.

2. Logic low sitting well below ground

What triggers it. A line whose low level is *negative* with respect to the probe's reference, with the whole signal still inside ±10 V.

What you see. The rate comes back wrong, in the same multiple-of-the-truth way as above.

How rare. Most of the negative range is fine: measured clean at **–0.1, –0.25, –0.5, –1.0 and –3.0 V** on a 3.3 V swing. Around **–2 V** some payloads fail. It is not a threshold — **–3 V works where –2 V does not** — so it is a band rather than a limit. Across a full seeded sweep with amplitude and offset both varied, **12 of 1 763** points were affected (**0.68 %**), and varying the amplitude alone over 0.5× to 1.6× caused **no failures at all**. Signals sitting entirely at or above 0 V are unaffected at any offset inside ±10 V.

Besides locking the rate: reference the probe to the line's own low level rather than to earth.

3. Long runs of one repeated byte, above 115 200 baud

What triggers it. All three of: a line faster than **115 200 baud**, a payload carrying **64 or more consecutive identical bytes**, and an unlucky capture placement. All three are needed.

Why the rate matters. The instrument digitises at 1 MS/s at most, and the app asks for eight samples per bit. Up to 115 200 it gets them; past that it cannot:

baud	samples per bit
9 600 – 76 800	8.33
115 200	**8.68 — fine, and it is the commonest fast rate**
128 000	7.81
153 600	**6.51 — where this is observed**
230 400	4.34
250 000	4.00

Why the payload matters. Inside a long run of one byte the wire carries only two gap lengths — nine bit times at one level and one at the other, over and over. Rate detection has nothing else to work from, and with six and a half samples per bit that is not enough to settle a bit time.

What you see. No decode at all — the byte area stays empty and no rate is reported. Unlike cases 1 and 2 this does not give you a wrong number; it declines, which is at least honest.

How rare. Measured over **2.7 million decodes** at eight capture placements each: **3.2 per million**, every one between 6.2 and 6.5 samples per bit, and never at 8 or more. Only 2 of the bench's 41 test waveforms produce it, and they are the two whose payloads carry 64- and 128-byte runs of a single value; a third waveform built the same way, but whose bytes never repeat consecutively, never produces it.

What to do — either of these is enough.

- **Lock the rate.** Type it into `Options ▸ Baud Rate`. **Verified on the exact failing capture: unlocked it failed 1 of 8 placements, locked it decoded 8 of 8.** Six and a half samples per bit is plenty to *decode* with; it is only the measurement that cannot be made.
- **If you control the link, run it at 115 200 or below** when you expect long runs of one byte — a flash dump reading back as solid `FF`, a zero-padded frame, protocol fill. Every standard rate from 9 600 to 115 200 gets at least 8.3 samples per bit and none of them shows this.

An idle line is **not** an instance of this. Idle is a continuous level with no framing at all, which the app handles separately; this needs a long run of framed bytes that happen to be equal.

Measured figures, the full tolerance envelope, press-by-press timings and the reasoning behind the design are in **REFERENCE.md**.